

MSDM 5056 Network Modeling  
Assignment 1 (due 24<sup>th</sup> September, 2025)

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

**(1) Adjacency matrix**

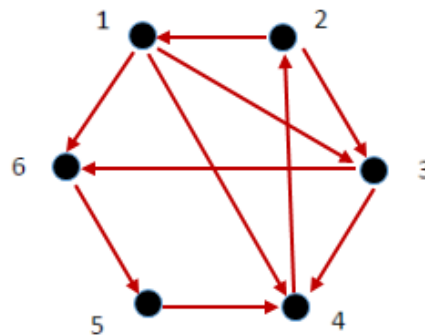
Consider the following adjacency matrix of a network.

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- (a) Is the network directed or undirected? (Explain why).
- (b) Draw the network.
- (c) What are the in-degree and out-degree of each node?
- (d) Determine the in-degree distribution and the out-degree distribution.

**Solution:**

- (a) The network is directed because the adjacency is asymmetric.
- (b)



- (c) Node 1 has in-degree 1 and out-degree 3. Node 2 has in-degree 1 and out-degree 2. Node 3 has in-degree 2 and out-degree 2. Node 4 has in-degree 3 and out-degree 1. Node 5 has in-degree 1 and out-degree 1. Node 6 has in-degree 2 and out-degree 1.
- (d) The in-degree distribution is given by  $P(0) = 0$ ,  $P(1) = 1/2$ ,  $P(2) = 1/3$ ,  $P(3) = 1/6$  and  $P(k) = 0$  for  $k > 3$ . The out-degree distribution is also given by  $P(0) = 0$ ,  $P(1) = 1/2$ ,  $P(2) = 1/3$ ,  $P(3) = 1/6$  and  $P(k) = 0$  for  $k > 3$ .

## (2) Diameter, eccentricity and radius

One can calculate the diameter  $D$  of certain types of networks exactly. Assume that each of the following networks has network size  $N$ .

- What is the diameter of a fully connected network?
- What is the diameter of a star network?
- What is the diameter of a linear chain (like Figure 1)?
- What is the diameter of a square portion of a square lattice with  $L$  nodes along each side (like Figure 2) in terms of  $L$ ?
- Show that the expression found in (d) approaches  $2\sqrt{N}$  in the limit  $N \gg 1$ .
- What is the diameter of a hypercubic lattice in  $d$  dimensions (e.g.  $d = 3$  corresponds to a cubic lattice) with  $L$  nodes along each side, in the limit  $L \gg 1$  (hence  $N \gg 1$ )?
- A Cayley tree is a symmetric regular tree in which each vertex is connected to the same number  $k$  of others until we get out to the leaves (Figure 3 uses  $k = 3$ ). Show that the number of vertices reachable in  $d$  steps from the central vertex is  $k(k-1)^{d-1}$  for  $d \geq 1$ . Hence find an expression for the diameter of a Cayley tree in terms of  $k$  and  $N$ .

- The **eccentricity** of a vertex  $v$  in a graph  $G$  is defined as

$$e(v) = \max\{\text{dist}(v, u) : u \in V(G)\}$$

while the **radius** of  $G$  is defined as

$$r(G) = \min\{e(v) : v \in V(G)\}$$

where  $V(G)$  denotes the set of vertices in  $G$ . Note that the diameter of  $G$  is equivalent to  $D(G) = \max\{e(v) : v \in V(G)\}$ . What is the eccentricity of each node of a linear chain (like Figure 1)? Hence, what is its radius?

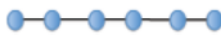


Figure 1

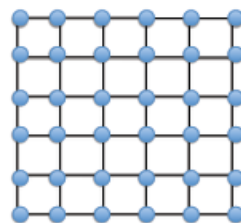


Figure 2

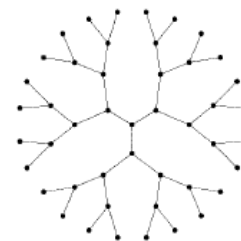


Figure 3

### Solution:

- Since a fully connected network is a network in which every node is linked to every other node, any two nodes are at distance  $d = 1$ . The diameter is therefore  $D = 1$ .
- A star network is a network in which a central node is linked to  $N - 1$  leaf nodes of degree 1. The diameter of the star network is therefore  $D = 2$ .

- (c) In a chain of  $N$  nodes the maximum distance is the distance between the two nodes at the edges of the chain. Therefore  $D = N - 1$ .
- (d) In a square portion of a square lattice, the most distant pair of points are the ones formed by the nodes at the two opposite corners of the square lattice. These nodes are at distance  $d = 2(L - 1)$ , so the diameter of the network is  $D = 2(L - 1)$ .
- (e) Since the number of nodes is  $N = L^2$ , we have  $D = 2(L - 1) = 2(\sqrt{N} - 1) \approx 2\sqrt{N}$  in the limit  $N \gg 1$ .
- (f) In a similar fashion, since  $L = \sqrt[d]{N}$ , we have  $D = d(L - 1) \approx d\sqrt[d]{N}$ .
- (g) In the first step, there are  $k$  vertices connected to the central vertex. In the next step, each of the  $k$  vertices can connect to  $(k - 1)$  vertices. In the third step, there will be  $(k - 1)$  vertices that connect to each of the  $k(k - 1)$  vertices, which gives  $k(k - 1)^2$  vertices. In the  $d$ th step, there will be  $k(k - 1)^{d-1}$  vertices in the outermost layer. The total number of vertices from the central vertex to the outermost layer is a sum

$$N = 1 + k \sum_{i=1}^d (k - 1)^{i-1} = 1 + \frac{k[(k - 1)^d - 1]}{(k - 1) - 1}$$

which gives

$$d = \frac{\ln[N(k - 1) + 1] - \ln k}{\ln(k - 1)}$$

The diameter is  $D = 2d$ .

When  $k = 2$ , the tree becomes a chain like Figure 1, and one can verify the answer by using e.g., L'Hopital's rule to get  $N = 1 + 2d$  or  $D = N - 1$ , which agrees with (c).

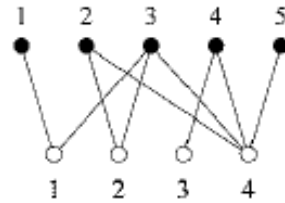
- (h) For a linear chain of  $N$  nodes, the eccentricity of the end points are  $N - 1$ . Denote an end point to be  $i = 0$ . The eccentricity of a node  $i$  is  $\max(i, N - 1 - i)$ . The radius is the eccentricity of the middle nodes. If  $N$  is odd, radius is  $(N - 1)/2$ . If  $N$  is even, radius is  $N/2$ .

### (3) Bipartite matrix

- (a) Consider a bipartite network with two types of vertices. Suppose that there are  $n_1$  vertices of type 1 and  $n_2$  vertices of type 2.
- What is the maximum number of links this network can have?
  - How many links of a non-bipartite and complete network of size  $n = n_1 + n_2$  cannot occur here?
  - Show that the mean degrees  $c_1$  and  $c_2$  of the two types of vertices are related by

$$c_2 = \frac{n_1}{n_2} c_1$$

(b) Consider the following undirected bipartite network.



Find the incidence matrix of this bipartite network. Draw the one-mode projections of the bipartite network.

**Solution:**

(a)

(i) The maximum number of links is  $n_1 \times n_2$ .

(ii) The maximum number of links for the non-bipartite and complete network is

$\frac{1}{2}(n_1 + n_2)(n_1 + n_2 - 1)$ . Therefore, the difference is  $\frac{1}{2}[n_1(n_1 - 1) + n_2(n_2 - 1)]$ .

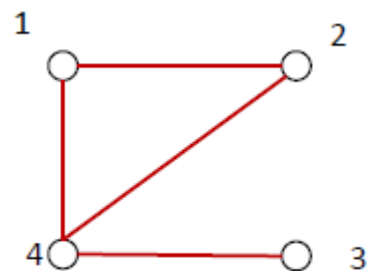
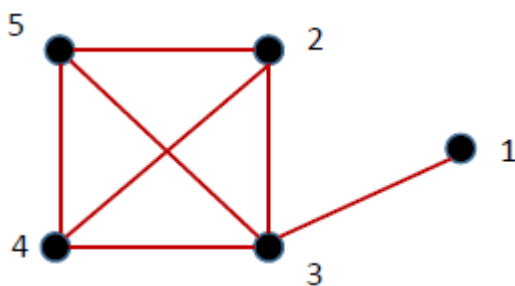
(iii) Since the total number of links of the two types of vertices are equal, i.e.,

$\sum_i k_i = n_1 c_1 = n_2 c_2$ , we have  $c_2 = \frac{n_1}{n_2} c_1$ .

(b) The incidence matrix  $B$  is

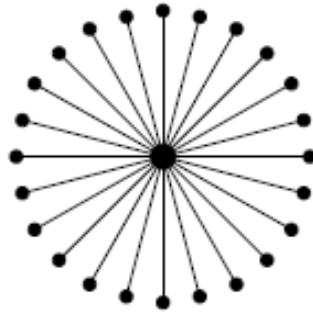
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The one-mode projections are shown below.



#### (4) Spectrum of star graphs

A general star graph consists of a central node with  $n - 1$  other nodes connected to it. The following example shows  $n = 25$ .



- (a) What is the average degree of a general star graph?
- (b) Find the eigenvalues of the adjacency matrix of a general star graph.
- (c) Now find the eigenvalues of the Laplacian of a general star graph. What do its eigenvalues tell you about it? (Refer to p.56 of Lecture 2)

**Solution:**

(a) The average degree is  $\frac{1}{n}[(n - 1) \times 1 + 1 \times (n - 1)] = 2 - \frac{2}{n}$ .

(b) The adjacency matrix takes the form

$$\begin{pmatrix} 0 & 1 & 1 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

The eigenvalues are obtained by finding the roots of the determinant

$$\begin{vmatrix} -\lambda & 1 & 1 & \cdots & 1 \\ 1 & -\lambda & 0 & \cdots & 0 \\ 1 & 0 & -\lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & -\lambda \end{vmatrix}$$

which give

$$\begin{aligned} \lambda^n - (n - 1)\lambda^{n-2} &= 0 \\ \Rightarrow \lambda^{n-2} [\lambda^2 - (n - 1)] &= 0 \end{aligned}$$

Therefore, the eigenvalues are  $\sqrt{n - 1}$ , 0 (with a multiplicity of  $n - 2$ ), and  $-\sqrt{n - 1}$ .

(c) The Laplacian takes the form

$$\begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -1 & 1 & 0 & \cdots & 0 \\ -1 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

Its eigenvalues are obtained by finding the roots of the determinant

$$\begin{vmatrix} n-1-\lambda & -1 & -1 & \cdots & -1 \\ -1 & 1-\lambda & 0 & \cdots & 0 \\ -1 & 0 & 1-\lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \cdots & 1-\lambda \end{vmatrix}$$

which gives

$$\begin{aligned} (n-1-\lambda)(1-\lambda)^{n-1} - (n-1)(1-\lambda)^{n-2} &= 0 \\ \Rightarrow (1-\lambda)^{n-2}\lambda(\lambda-n) &= 0 \end{aligned}$$

Therefore, the eigenvalues are  $n$ ,  $1$  (with a multiplicity of  $n-2$ ), and  $0$ .

The second smallest eigenvalue is equal to  $1$ , which is larger than zero. This means the graph is a connected graph, consisting of only one single component. Notice that all the eigenvalues are larger than or equal to  $0$ . There is only one zero eigenvalue, again meaning that there is only one component in the graph.